Introduction

Simpson's non abelian Hodge correspondence establishes a homeomorphism between the moduli space of semistable Higgs bundles on a smooth projective curve X and the character variety of the curve. This can be understood as relating the geometry of the generic fiber and the special fiber inside the moduli space of λ-connections. Namely, for a complex parameter λ and a vector bundle $\mathcal{E}$ on X, define a λ-connection on $\mathcal{E}$ to be a $\mathbb{C}$-linear operator

$$\nabla \colon \mathcal{E} \to \mathcal{E} \otimes \Omega_X$$

satisfying the Leibniz rule up to the scalar λ. Explicitly, given sections $f \in \mathcal{O}_X(U)$ and $s \in \mathcal{E}(U)$ we have

$$\nabla(f\,s) = f\,\nabla(s) + \lambda\, s \otimes \mathrm{d}f.$$

The Hodge moduli space $\mathcal{M}^0_{n,Hdg}(X)$ parametrizes triples $(\mathcal{E}, \lambda, \nabla)$, where $\mathcal{E}$ is a rank n and degree 0 vector bundle on X and ∇ is a λ-connection on $\mathcal{E}$, subject to a stability condition. The assignment

$$(\mathcal{E}, \lambda, \nabla) \mapsto \lambda$$

gives $\mathcal{M}^0_{n,Hdg}(X)$ the structure of a family over $\mathbb{A}^1$, trivial over $\mathbb{G}_m$. Notice that the generic fiber (identified with $\lambda = 1$) is the moduli space of vector bundles on X with a (flat) connection, i.e. the De Rham moduli space $\mathcal{M}^0_{n,DR}(X)$. The special fiber is the moduli space of semistable Higgs bundles on X, i.e. the Dolbeault moduli space $\mathcal{M}^0_{n,Dol}(X)$.

Notice moreover that the De Rham moduli space is complex-analytically isomorphic to the moduli space of rank n local systems on X, which can be identified with the character variety

$$\mathcal{M}^0_{n,Betti}(X) := [\mathrm{Hom}(\pi_1(X), GL_n)/GL_n]\,.$$

In short, we have the following picture

$$\begin{tikzcd}
\mathcal{M}^0_{n,Dol}(X) \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{M}^0_{n,Hdg}(X) \arrow[d] & \mathcal{M}^0_{n,DR}(X) \arrow[l, hook'] \arrow[r, "\cong"] \arrow[d] \arrow[dl, phantom, "\llcorner"] & \mathcal{M}^0_{n,B}(X) \\
\{0\} \arrow[r] & \mathbb{A}^1 & \{1\} \arrow[l] &
\end{tikzcd} \tag{0.1}$$

and Simpson's correspondence shows that the analytic spaces underlying $\mathcal{M}^0_{n,Dol}(X)$ and $\mathcal{M}^0_{n,DR}(X)$ are homeomorphic. From an algebraic perspective, the main consequence is that these varieties have isomorphic cohomology rings. Indeed, we get that the inclusions of the fibers in the total space induce isomorphisms of cohomology rings

$$H^*(\mathcal{M}^0_{n,Dol}(X)) \cong H^*(\mathcal{M}^0_{n,Hdg}(X)) \cong H^*(\mathcal{M}^0_{n,DR}(X));$$

and that the analytic isomorphism $\mathcal{M}^0_{n,DR}(X) \cong \mathcal{M}^0_{n,B}(X)$ induces an isomorphism

$$H^*(\mathcal{M}^0_{n,DR}(X)) \cong H^*(\mathcal{M}^0_{n,B}(X)).$$

Notice that so far we have described only moduli spaces of degree 0 vector bundles. Indeed, for a vector bundle to admit a flat connection it needs to have vanishing Chern classes, so the description given of $\mathcal{M}_{DR}$ only makes sense in degree 0. However, the moduli space of Higgs bundles $\mathcal{M}_{Dol}$ can be constructed easily for any degree d. Classically, this issue has been solved by replacing the De Rham and Betti moduli spaces with *twisted* versions. Namely $\mathcal{M}^d_{n,DR}(X)$ is the moduli space of semistable vector bundles with a logarithmic connection with a single pole $p \in X$ of multiplicity at most n and residue $-\frac{2\pi i d}{n}$, and

$$\mathcal{M}^d_{n,B}(X) = \left[\left\{ (A_i, B_i)_{i=1}^g \in GL_n^{2g} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = \mathrm{e}^{-\frac{2\pi \mathrm{i}}{n} d} Id \right\} / GL_n \right]$$

is the *twisted character variety*.

The main purpose of this article is to give an alternative description of $\mathcal{M}^d_{n,DR}(X)$ and $\mathcal{M}^d_{n,B}(X)$, which more clearly fits into a picture as the one described in (0.1).

In order to do this, the main insight is that the meromorphic structure around the point $p \in X$ can be interpreted as to be holomorphic around an orbifold point and similarly the twist appearing in $\mathcal{M}^d_{n,B}(X)$ can be interpreted as monodromy around an orbifold point. Algebraically, the way to realize this orbifold structure is to construct a stacky curve $\mathcal{X}$ as the n-th root stack of X at the point p, i.e. as

$$\begin{tikzcd}
\mathcal{X} = \sqrt[n]{p/X} \arrow[r] \arrow[d, "\pi"'] \arrow[dr, phantom, "\lrcorner"] & {[\mathbb{A}^1/\mathbb{G}_m]} \arrow[d, "\wedge n"] \\
X \arrow[r, "{(\mathcal{O}(p), s_p)}"'] & {[\mathbb{A}^1/\mathbb{G}_m]}.
\end{tikzcd}$$

In order to get back the picture in (0.1), we need to reduce to the study of degree 0 vector bundles on $\mathcal{X}$. Indeed, the main results presented are about realizing moduli spaces of degree d vector bundles on X as moduli spaces of degree 0 vector bundles on $\mathcal{X}$ and about realizing $\mathcal{M}^d_{n,B}(X)$ as moduli space of representations of $\pi_1(\mathcal{X})$. In what follows we give a more detailed description of how this insight is formalized.

Vector bundles on $\mathcal{X}$ have a third numerical invariant beyond rank and degree, namely the representation obtained as stalk at the stacky point. We denote by $w(-d)$ the weight $-d$ character of μ_n and by $n \cdot w(-d)$ the corresponding n-dimensional representation. Then the main result comparing moduli spaces on X and on $\mathcal{X}$ is the following.

Theorem 0.1. The twisted pullback and pushforward along the moduli map $\mathcal{X} \to X$ define equivalences of algebraic stacks

$$\mathcal{M}^d_{n,Dol}(X) \cong \mathcal{M}^0_{n \cdot w(-d),Dol}(\mathcal{X}).$$

Passing to the study of the Betti moduli space, a first observation is that the replacement of p by an orbifold point changes the fundamental group of X by adding a cyclic monodromy element.

Namely, we get

$$\pi_1(\mathcal{X}) = \left\langle a_i,\, b_i,\, c \;\middle|\; \prod_{i=1}^{g}[a_i, b_i] = c,\; c^n = 1 \right\rangle.$$

Then, we get the twisted character variety thanks to the following theorem.

Theorem 0.2. The connected component of the Betti moduli stack of $\mathcal{X}$ corresponding to the numerical invariant $n \cdot w(-d)$ is isomorphic to the twisted Betti moduli space of X, i.e. we have

$$\mathcal{M}^0_{n\cdot w(-d),B}(\mathcal{X}) = \left[\left\{(A_i, B_i)_{i=1}^g \in GL_n^{2g} \;\middle|\; \prod_{i=1}^{g}[A_i, B_i] = \mathrm{e}^{-\frac{2\pi \mathrm{i}}{n}d} Id\right\}/GL_n\right].$$

Just like in the case described in (0.1), these moduli spaces can be realized as the special and the generic fiber inside the Hodge moduli space, i.e. we have

$$\begin{tikzcd}
\mathcal{M}^0_{n\cdot w(-d),Dol}(\mathcal{X}) \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{M}^0_{n\cdot w(-d),Hdg}(\mathcal{X}) \arrow[d] & \mathcal{M}^0_{n\cdot w(-d),DR}(\mathcal{X}) \arrow[l, hook'] \arrow[r, "\cong"] \arrow[d] \arrow[dl, phantom, "\llcorner"] & \mathcal{M}^0_{n\cdot w(-d),B}(\mathcal{X}) \\
\{0\} \arrow[r] & \mathbb{A}^1 & \{1\}, \arrow[l] &
\end{tikzcd}$$

where the isomorphism

$$\mathcal{M}^0_{n\cdot w(-d),DR}(\mathcal{X}) \xrightarrow{\cong} \mathcal{M}^0_{n\cdot w(-d),B}(\mathcal{X})$$

is complex analytic but not algebraic. As before, we get that the inclusions of the fibers in the total space induce isomorphisms of cohomology rings

$$H^*(\mathcal{M}^0_{n\cdot w(-d),Dol}(\mathcal{X})) \cong H^*(\mathcal{M}^0_{n\cdot w(-d),Hdg}(\mathcal{X})) \cong H^*(\mathcal{M}^0_{n\cdot w(-d),DR}(\mathcal{X}));$$

and that the analytic isomorphism $\mathcal{M}^0_{n\cdot w(-d),DR}(\mathcal{X}) \cong \mathcal{M}^0_{n\cdot w(-d),B}(\mathcal{X})$ induces an isomorphism

$$H^*(\mathcal{M}^0_{n\cdot w(-d),DR}(\mathcal{X})) \cong H^*(\mathcal{M}^0_{n\cdot w(-d),B}(\mathcal{X})).$$

This approach has the further advantage of simplifying the calculation of the Hodge weights of the tautological classes on $\mathcal{M}^d_{n,B}(X)$. In [She17], Shende computed the weights of the tautological classes on the character variety of a topological space by replacing the space by a triangulation and then realizing this triangulation as a simplicial scheme. He later employed the approach presented in [HR08, §2.2] to compute compute the weights also in the case of twisted character varieties, by reducing it to the case of PGL_n-character varieties. The following is the result of this computation.

Theorem 0.3 (Shende)**.** Under the isomorphism $H^*(\mathcal{M}^0_{n,DR}(X)) \cong H^*(\mathcal{M}^0_{n,Dol}(X))$, the Künneth components $e_{i,j}$ of the i-th Chern class of the universal bundle $c_i(\mathcal{E}_{\mathrm{univ}})$ are mapped to cohomology classes of weight $2i$. The same holds true in the twisted case, under the isomorphism $H^*(\mathcal{M}^d_{n,DR}(X)) \cong H^*(\mathcal{M}^d_{n,Dol}(X))$.

Our approach gives an alternative proof for the twisted case, which doesn't need to use the case of PGL_n-bundles and the decomposition presented by Hausel and Rodriguez-Villegas in [HR08]. Indeed, using Theorem 0.2 we see that the twisted character variety can be realized as an actual variety of local systems on $\mathcal{X}$. Therefore Shende's argument can be extend directly to cover this case. The main ingredient is the construction of a simplicial scheme $\Delta_{\mathcal{X}}$ which realizes a triangulation of the stacky curve $\mathcal{X}$. In the article we explain in detail the construction of this simplicial decomposition. Once this is done, the argument amounts to using the equivalence

$$\mathcal{M}^0_{n,B}(\mathcal{X}) = [\mathrm{Hom}(\pi_1(\mathcal{X}), GL_n)/GL_n] \cong \mathcal{H}om_{SSch}(\Delta_{\mathcal{X}}, BGL_n)$$

and the universal object on $\mathcal{H}om_{SSch}(\Delta_{\mathcal{X}}, BGL_n)$ to carry out the computation of the weights. Using this argument we get the following theorem.

Theorem 0.4. The cohomology classes on on $\mathcal{M}^0_{n,B}(\mathcal{X})$ obtained by pulling back the i-th tautological classes on $\mathcal{M}^0_{n,DR}(\mathcal{X})$ have pure weight $2i$.

By restricting to the $n \cdot w(-d)$-component of the Betti moduli space we get an alternative proof of Theorem 0.3.

1 Vector bundles on the stacky curve

Let X be a smooth, projective, genus g curve over $\mathbb{C}$ and let $p \in X$ be a closed point. Following [Cad07], we construct $\mathcal{X} = \sqrt[n]{p/X}$ as a root stack over X, with multiplicity n in p.

Consider p as an effective Cartier divisor on X. Associated to it, there is a pair $(\mathcal{O}_X(p), s_p)$, where $\mathcal{O}_X(p)$ is a line bundle and s_p is a section of $\mathcal{O}_X(p)$ vanishing at p, called the *tautological section*. This pair defines a morphism

$$X \xrightarrow{(\mathcal{O}(p),\, s_p)} [\mathbb{A}^1/\mathbb{G}_m].$$

Let $\wedge n \colon [\mathbb{A}^1/\mathbb{G}_m] \to [\mathbb{A}^1/\mathbb{G}_m]$ be the morphism induced by taking the n-th power maps on $\mathbb{A}^1$ and on $\mathbb{G}_m$. Define $\mathcal{X}$ as the pullback

$$\begin{tikzcd} \mathcal{X} \arrow[r] \arrow[d, "\pi"'] \arrow[dr, phantom, "\lrcorner", very near start] & {[\mathbb{A}^1/\mathbb{G}_m]} \arrow[d, "\wedge n"] \\ X \arrow[r, "(\mathcal{O}(p),\, s_p)"'] & {[\mathbb{A}^1/\mathbb{G}_m]}. \end{tikzcd} \tag{1.1}$$

The vertical map π is a coarse moduli map, see [Cad07, Corollary 2.3.7]. The stack $\mathcal{X}$ comes equipped with a pair $(\mathcal{O}(\frac{1}{n}p), \frac{1}{n}s_p)$, where $\mathcal{O}(\frac{1}{n}p)$ is a line bundle and $\frac{1}{n}s_p$ is a section of $\mathcal{O}(\frac{1}{n}p)$. This pair corresponds to the upper horizontal morphism in (1.1), and is characterized by the properties

$$\pi^*\mathcal{O}(p) = \mathcal{O}(\tfrac{1}{n}p)^{\otimes n}, \quad \pi^* s_p = (\tfrac{1}{n}s_p)^{\otimes n}. \tag{1.2}$$

The sheaf $\mathcal{O}(\frac{1}{n}p)$ is called the *tautological sheaf* of the stacky curve $\mathcal{X}$. It corresponds to the composition

$$\mathcal{X} \to [\mathbb{A}^1/\mathbb{G}_m] \to B\mathbb{G}_m.$$

With an abuse of notation, we denote by p also the stacky point in $\mathcal{X}$ above p. Notice that in p the stack $\mathcal{X}$ has the structure

$$\begin{tikzcd} B\mu_n \arrow[r, hook] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{X} \arrow[d, "\pi"] \\ \{p\} \arrow[r, hook] & X. \end{tikzcd}$$

More explicitly, if $U \cong \mathbb{A}^1$ is an affine neighborhood of p, we have that

$$\mathcal{X}|_U \cong [\mathbb{A}^1/\mu_n],$$

see [Cad07, Lemma 2.3.1]. Away from p the morphism π is an isomorphism

$$\mathcal{X} \setminus \{p\} \cong X \setminus \{p\}. \tag{1.3}$$

The goal of the construction is to identify vector bundles on X with their pullback along π. In particular, we need to mention the main properties of the pullback and pushforward functors π^* and π_*.

Proposition 1.1. [AV02] The pullback and pusforward functors along π have the following properties.

- The pushforward $\pi_* \colon \mathrm{QCoh}(\mathcal{X}) \to \mathrm{QCoh}(X)$ is exact and restricts to a functor $\pi_* \colon \mathcal{B}un(\mathcal{X}) \to \mathcal{B}un(X)$. [AV02, Theorem 2.3.4]
- The pullback $\pi_* \colon \mathrm{QCoh}(X) \to \mathrm{QCoh}(\mathcal{X})$ is exact. [Taa24, Proposition 1.2.3].
- The natural morphism $\mathcal{O}_X \to \pi_*\mathcal{O}_\mathcal{X}$ is an isomorphism. [Taa24, Theorem 1.2.4]
- The natural morphism $\mathrm{Hom}_\mathcal{X}(\mathcal{O}_\mathcal{X}, \pi^*\pi_*\mathcal{F}) \to \mathrm{Hom}_\mathcal{X}(\mathcal{O}_\mathcal{X}, \mathcal{F})$ is an isomorphism. [Taa24, Theorem 1.2.4]
- There is a natural isomorphism $\hom_X(\mathcal{O}_X, \pi_*\mathcal{F}) \to \hom_\mathcal{X}(\mathcal{O}_\mathcal{X}, \mathcal{F})$. In particular, $H^i(X, \pi_*\mathcal{F}) = H^i(\mathcal{X}, \mathcal{F})$. [Taa24, Theorem 1.2.4]

We now want to understand the numerical invariants that decompose the moduli space of vector bundles on $\mathcal{X}$ into connected components. Let $\mathcal{E}$ be a rank m vector bundle on $\mathcal{X}$. By pulling back along the inclusion of the stacky point

$$i \colon B\mu_n \hookrightarrow \mathcal{X}$$

we get a vector bundle on $B\mu_n$, i.e. an m-dimensional representation of the cyclic group μ_n. Since μ_n is an abelian group, the representation decomposes as a direct sum of characters. In particular, denoting by $w(i)$ the weight i character, we get a decomposition

$$i^*\mathcal{E} = \bigoplus_{i \in \mathbb{Z}/n\mathbb{Z}} w(i)^{m_i}.$$

Definition 1.2. [Taa24, Definition 1.2.7] To keep track of the multiplicities of all of the weights appearing, we denote the multiplicity vector of $\mathcal{E}$ at p by

$$\underline{m}(\mathcal{E}) := m_0\, w(0) + \cdots + m_{n-1}\, w(n-1).$$

Notice that the rank is immediately recovered from the multiplicity vector as $m = m_0 + \cdots + m_{n-1}$.

By construction the tautological line bundle $\mathcal{O}(\frac{1}{n}p)$ has weight 1 in p, i.e. we have

$$\underline{m}(\mathcal{O}(\tfrac{1}{n}p)) = 1 \cdot w(1).$$

Remark 1.3. [Taa24, §1.2] Since pullback commutes with tensor product and $w(i) \otimes w(j) = w(i+j)$, we see that tensoring with the tautological line bundle acts as a shift operator on the multiplicity vector, i.e. mapping $w(i)$ to $w(i+1)$.

Proposition 1.4. [Taa24, Example 1.2.9] If F is a coherent sheaf on X, then $m_p(\pi^* F) = n \cdot w(0)$. Namely, the pullback $\pi^* F$ is endowed with the trivial μ_n action at p.

The other numerical invariant associated to a vector bundle is its degree. We need to define a notion of degree for vector bundles on $\mathcal{X}$, respecting the classical formulas for the degree of the direct sum and tensor product of two vector bundles. Keep in mind that we want to identify vector bundles on X with their pullback along π, so we want

$$\pi^* \colon \mathcal{B}un_n(X) \to \mathcal{B}un_n(\mathcal{X})$$

to be degree preserving. It turns out that this property completely determines the definition of the degree.

Definition 1.5. (See [Taa24, Definition 1.2.9]). Let $K(\mathcal{X})$ denote the Groethendieck group of the stacky curve $\mathcal{X}$. Then the degree is defined to be the unique group homomorphism

$$\deg \colon K(\mathcal{X}) \to \mathbb{Q}$$

such that the pullback $\pi^* \colon K(X) \to K(\mathcal{X})$ is degree preserving.

See [Taa24, §1.2] for a more explicit definition. From Equation (1.2) we get

$$\deg(\mathcal{O}(\tfrac{1}{n}p)) = \tfrac{1}{n},$$

which further justifies the notation $\mathcal{O}(\frac{1}{n}p)$.

In practice, we have an explicit formula to relate the degree of a vector bundle to the degree of its pushforward along π.

Proposition 1.6. [Taa24, Theorem 1.2.30] Let $\mathcal{E}$ be a vector bundle on $\mathcal{X}$ and suppose $\underline{m}(\mathcal{E}) = m_0\, w(0) + \cdots + m_{n-1}\, w(n-1)$. Then we have

$$\deg \mathcal{E} = \deg(\pi_* \mathcal{E}) + \sum_{i=0}^{n-1} i\, m_i.$$

Theorem 1.7. [Taa24, Corollary 2.0.9, Theorem 2.0.10] For a fixed multiplicity vector $\underline{m} = m_0\, w(0) + \cdots + m_{n-1}\, w(n-1)$ and a degree $d \in \frac{1}{n}\mathbb{Z}$, denote by $\mathcal{B}un^d_{\underline{m}}(\mathcal{X})$ the substack of $\mathcal{B}un(\mathcal{X})$ defined by

$$\mathcal{B}un^d_{\underline{m}}(\mathcal{X}) := \langle \mathcal{E} \in \mathcal{B}un(\mathcal{X}) \mid (\underline{m}(\mathcal{E}), \deg \mathcal{E}) = (\underline{m}, d) \rangle .$$

Then the substacks $\mathcal{B}un^d_{\underline{m}}(\mathcal{X})$ are the connected components of $\mathcal{B}un(\mathcal{X})$. Moreover, the stack $\mathcal{B}un(\mathcal{X})$ is smooth, so the connected components coincide with the irreducible components and they are smooth.

Remark 1.8. In [Taa24, §2] the connected components of $\mathcal{B}un(\mathcal{X})$ are described by a different pair of numerical invariants, namely the rank and the *twisted degrees* vector, defined by

$$d_i(\mathcal{E}) := \deg\left(\pi_*\mathcal{E} \otimes \mathcal{O}(\tfrac{i}{e_p}p)\right)$$

for $i = 0, \dots, n-1$. But the pair $(\operatorname{rk}\mathcal{E}, \underline{d}(\mathcal{E}))$ is equivalent to the pair $(m_p(\mathcal{E}), \deg\mathcal{E})$, as one can easily see using [Taa24, Corollary 1.2.21]. Therefore Theorem 1.7 is indeed the same as [Taa24, Corollary 2.0.9, Theorem 2.0.10].

We are finally ready to compare the moduli spaces of vector bundles on X with moduli spaces of vector bundles on $\mathcal{X}$.

Proposition 1.9. The pullback and pushforward functors along the moduli map $\pi\colon \mathcal{X} \to X$ define inverse equivalences of algebraic stacks

$$\begin{tikzcd} \mathcal{B}un_n^d(X) \arrow[r, bend left, "\pi^*"] \arrow[r, phantom, "\cong"] & \mathcal{B}un_{n\cdot w(0)}^d(\mathcal{X}), \arrow[l, bend left, "\pi_*"] \end{tikzcd}$$

which preserve (semi)stability.

Proof. First we need to observe that the functors are well defined between the given spaces. By Proposition 1.1 both functors map vector bundles to vector bundles, so wee need to check that they respect the given invariants.

For π^*, the degree on $\mathcal{X}$ was defined so that π^* would preserve the degree. Moreover, we noticed in Proposition 1.4 that for $F \in \mathcal{B}un_n(X)$, the multiplicity of π^*F at p is $n \cdot w(0)$, or equivalently that $\pi^*F|_{B\mu_n}$ is the trivial representation. Therefore π^* restricts as a well defined functor

$$\pi^*\colon \mathcal{B}un_n^d(X) \to \mathcal{B}un_{n\cdot w(0)}^d(\mathcal{X}).$$

In the other direction, notice that the rank of a vector bundle on a connected stack is the same at each point and π is an isomorphism away from p. Therefore π_* maps rank n vector bundles to rank n vector bundles. Finally, Proposition 1.6 shows that, if the multiplicities vector is $n \cdot w(0)$, then the degree of $\pi_*\mathcal{E}$ equals the degree of $\mathcal{E}$. Therefore, π_* restricts to a well defined functor

$$\pi_*\colon \mathcal{B}un_{n\cdot w(0)}^d(\mathcal{X}) \to \mathcal{B}un_n^d(X).$$

The fact that these are inverse equivalences are just [Taa24, Corollary 1.2.6, Theorem 1.2.10].

Notice that these are equivalences of categories of vector bundles, preserving both the rank and the degree. Therefore, they preserve (semi)stability. □

On $\mathcal{X}$ we can twist with the tautological line bundle to change the degree of a vector bundle. In this way we achieve the goal of realizing degree d vector bundles on X as degree 0 vector bundles on $\mathcal{X}$.

Theorem 1.10. The *twisted* pullback and pushforward functors, defined by

$$\mathcal{B}un_n^d(X) \underset{\pi_*(\mathcal{O}(-\frac{d}{n}p)\otimes -)}{\overset{\pi^*(-)\otimes\mathcal{O}(-\frac{d}{n}p)}{\rightleftarrows}} \mathcal{B}un^0_{n\cdot w(-d)}(\mathcal{X})$$

define inverse equivalences of algebraic stacks. These equivalences preserve (semi)stability, hence they restrict to equivalences

$$\mathcal{B}un_n^{d,ss}(X) \underset{\pi_*(\mathcal{O}(-\frac{d}{n}p)\otimes -)}{\overset{\pi^*(-)\otimes\mathcal{O}(-\frac{d}{n}p)}{\rightleftarrows}} \mathcal{B}un^{0,ss}_{n\cdot w(-d)}(\mathcal{X}).$$

Proof. Notice that tensoring with $\mathcal{O}(-\frac{1}{n}p)^{\otimes d}$ defines an equivalence $\mathcal{B}un^d_{n\cdot w(0)}(\mathcal{X}) \cong \mathcal{B}un^0_{n\cdot w(-d)}(\mathcal{X})$. Explicitly,

$$\begin{tikzcd} \mathcal{B}un^d_{n\cdot w(0)}(\mathcal{X}) \arrow[r, bend left, "-\otimes\mathcal{O}(-\frac{d}{n}p)"] \arrow[r, phantom, "\cong"] & \mathcal{B}un^0_{n\cdot w(-d)}(\mathcal{X}). \arrow[l, bend left, "\mathcal{O}(\frac{d}{n}p)\otimes -"] \end{tikzcd}$$

Clearly the two maps are inverse to each other. The only nontrivial thing to show is that they respect the defined invariants. Firstly, it was already observed in Remark 1.3 that the tautological line bundle acts as a shift operator on the multiplicities. Therefore, $w(0)$ becomes $w(-d)$ after tensoring and vice versa. Secondly, the formula for the degree of a tensor product gives

$$\deg\left(\mathcal{E}\otimes\mathcal{O}(-\tfrac{d}{n}p)\right) = \deg\mathcal{E} + n\left(-\frac{d}{n}\right) = 0.$$

Finally, notice how these equivalences also preserve (semi)stability. Indeed, let $\mathcal{E}\in\mathcal{B}un^d_{n\cdot w(0)}(\mathcal{X})$ be a semistable vector bundle. Since the indicated functors are equivalences, in particular they preserve subbundles. Therefore, it suffices to show that if $\mathcal{E}'\subseteq\mathcal{E}$ is a subbundle such that $\mu(\mathcal{E}')\leq\mu(\mathcal{E})$, then $\mu(\mathcal{E}'\otimes\mathcal{O}(-\frac{d}{n}p))\leq\mu(\mathcal{E}\otimes\mathcal{O}(-\frac{d}{n}p))$. This follows from the simple computation

$$\mu(\mathcal{E}'\otimes\mathcal{O}(-\tfrac{d}{n}p)) = \frac{\deg\mathcal{E}' - \operatorname{rk}\mathcal{E}'\frac{d}{n}}{\operatorname{rk}\mathcal{E}'} = \mu(\mathcal{E}') - \frac{d}{n}.$$

Repeating the same computation for $\mathcal{E}$, one shows that $\mu(\mathcal{E}'\otimes\mathcal{O}(-\frac{d}{n}p))\leq\mu(\mathcal{E}\otimes\mathcal{O}(-\frac{d}{n}p))$. The computation in the other direction is completely analogous.

The thesis follows after composing with the equivalences described in Proposition 1.9. □

2 Higgs bundles and Dolbeault moduli space

Recall that the end goal is to study the non abelian Hodge correspondence in degree 0 on $\mathcal{X}$ to get information about the non abelian Hodge correspondence in degree d on X. For this reason, we need to elevate Theorem 1.10 to an equivalence of stacks of Higgs bundles.

A first step in this direction is to compare the cotangent sheaf on X and on $\mathcal{X}$.

Theorem 2.1. [Taa24, Theorem 1.2.34] The cotangent sheaf on $\mathcal{X}$ is

$$\Omega_{\mathcal{X}} \cong \pi^*\Omega_X \otimes \mathcal{O}(\tfrac{1}{n}p)^{\otimes n-1}.$$

In order to extend Theorem 1.10 to Higgs bundles, we need to lift the functors π^* and π_* to the Higgs fields. A Higgs bundle on X is a pair (E, φ) where E is a vector bundle on X and φ is an $\mathcal{O}_X$-linear operator

$$\varphi\colon E \to E \otimes \Omega_X,$$

called the *Higgs field*. After pulling back, we get an operator

$$\pi^*\varphi\colon \pi^*E \to \pi^*E \otimes \pi^*\Omega_X.$$

In order to get a Higgs field on π^*E we compose with the canonical morphism $\pi^*\Omega_X \to \Omega_{\mathcal{X}}$. In the case under study, we have an explicit expression for this canonical morphism.

Lemma 2.2. Under the isomorphism

$$\Omega_{\mathcal{X}} \cong \pi^*\Omega_X \otimes \mathcal{O}(\tfrac{1}{n}p)^{\otimes n-1}$$

described in Theorem 2.1, the canonical morphism

$$\pi^*\Omega_X \to \Omega_{\mathcal{X}} \cong \pi^*\Omega_X \otimes \mathcal{O}(\tfrac{n-1}{n}p)$$

has the form $n\,\mathrm{id}_{\pi^*\Omega_X} \otimes s^{n-1}$, where s is the tautological section of $\mathcal{O}(\frac{1}{n}p)$.

Proof. The coarse moduli map $\pi\colon \mathcal{X} \to X$ locally around p has the following form

$$\begin{tikzcd}
\mathbb{A}^1 = \operatorname{Spec} k[x] \arrow[d, "q'"] \arrow[ddr, bend left, "q=\pi\circ q'"', "x\mapsto t=x^n"] & \\
\mathcal{X} = [\mathbb{A}^1/\mu_n] \arrow[d, "\pi"] & \\
X = \mathbb{A}^1/\mu_n \arrow[r, equal] & \operatorname{Spec} k[x^n] = \operatorname{Spec} k[t].
\end{tikzcd}$$

Notice that we have the relation

$$\mathrm{d}t = \mathrm{d}x^n = n\,x^{n-1}\,\mathrm{d}x,$$

therefore the canonical morphism $q^*\Omega_X \to \Omega_{\mathbb{A}^1}$ corresponds to

$$\begin{aligned} k[x] \otimes_{k[t]} k[t]\,\mathrm{d}t &\to k[x]\,\mathrm{d}x \\ f(x) \otimes \mathrm{d}t &\mapsto f(x)\,n\,x^{n-1}\,\mathrm{d}x. \end{aligned}$$

Since q' is étale, it induces isomorphisms on the cotangent sheaves. Moreover, the section x corresponds via q' to the tautological section s of $\mathcal{O}(\frac{1}{n}p)$. In conclusion, we get that the canonical morphism $\pi^*\Omega_X \to \Omega_{\mathcal{X}} \cong \pi^*\Omega_X \otimes \mathcal{O}(\frac{n-1}{n}p)$ has the indicated form. $\square$

Now we can use this morphism to extend the pullback functor π^* to Higgs bundles. Define

$$\begin{aligned}(\pi^*, \pi^\#)\colon \operatorname{Higgs}_n^d(X) &\to \operatorname{Higgs}_{n\cdot w(0)}^d(\mathcal{X})\\ (E, \varphi) &\mapsto (\pi^* E, \pi^\# \varphi),\end{aligned}$$

where $\pi^\#\varphi$ is defined as the composition

$$\begin{aligned}\pi^\#\varphi\colon \pi^* E \xrightarrow{\pi^*\varphi} \pi^*(E\otimes\Omega_X) = \pi^* E\otimes\pi^*\Omega_X \\ \xrightarrow{\mathrm{id}\otimes n\,\mathrm{id}\otimes s^{n-1}} \pi^* E\otimes\pi^*\Omega_X\otimes\mathcal{O}(\tfrac{n-1}{n}p) = \pi^* E\otimes\Omega_{\mathcal{X}}.\end{aligned}$$

In the other direction, given $(\mathcal{E}, \eta) \in \operatorname{Higgs}_{n\cdot w(0)}^d(\mathcal{X})$, we want to use η to define a Higgs field on $\pi_*\mathcal{E}$. Using Proposition 1.9, we get that $\mathcal{E} = \pi^* E$, where $E \cong \pi_*\mathcal{E}$. By definition, η is a morphism in $\operatorname{Hom}_{\mathcal{X}}(\mathcal{E}, \mathcal{E}\otimes\Omega_{\mathcal{X}})$. Now, using Theorem 2.1, we find a chain of natural isomorphisms

$$\begin{aligned}\operatorname{Hom}_{\mathcal{X}}(\mathcal{E},\, \mathcal{E}\otimes\Omega_{\mathcal{X}}) &\cong \operatorname{Hom}_{\mathcal{X}}\left(\pi^* E,\, \pi^* E\otimes\pi^*\Omega_X\otimes\mathcal{O}(\tfrac{n-1}{n}p)\right)\\ &\cong \operatorname{Hom}_X\left(E,\, \pi_*(\pi^*(E\otimes\Omega_{\mathcal{X}})\otimes\mathcal{O}(\tfrac{n-1}{n}p))\right) &(2.1)\\ &\cong \operatorname{Hom}_X\left(E,\, E\otimes\Omega_X\otimes\pi_*\mathcal{O}(\tfrac{n-1}{n}p)\right) &(2.2)\\ &\cong \operatorname{Hom}_X\left(E,\, E\otimes\Omega_X\right), &(2.3)\end{aligned}$$

where (2.1) follows by the adjunction $\pi^* \dashv \pi_*$, (2.2) by the projection formula and (2.3) by the isomorphism $\pi_*\mathcal{O}(\frac{n-1}{n}p) \cong \mathcal{O}_X$ obtained from [Taa24, Corollary 1.2.13].
Composing the isomorphisms, we get

$$\begin{aligned}\operatorname{Hom}_X(E, E\otimes\Omega_X) &\xrightarrow{\cong} \operatorname{Hom}_{\mathcal{X}}(\mathcal{E}, \mathcal{E}\otimes\Omega_{\mathcal{X}})\\ \psi &\mapsto n\,\pi^*\psi\otimes s^{n-1}.\end{aligned}$$

This shows that there exists a unique $\psi \in \operatorname{Hom}_X(E, E\otimes\Omega_X)$ such that $\eta = n\,\pi^*\psi\otimes s^{n-1}$, where in particular we have $\psi = \frac{1}{n}\,\pi_*\eta$. Therefore, we can define

$$\pi_\#\eta := \frac{1}{n}\,\pi_*\eta.$$

This gives a well defined functor $(\pi_*, \pi_\#)$.

$$(\pi_*, \pi_\#)\colon \operatorname{Higgs}_{n\cdot w(0)}^d(\mathcal{X}) \to \operatorname{Higgs}_n^d(X)$$

Proposition 2.3. The functors $(\pi^*, \pi^\#)$ and $(\pi_*, \pi_\#)$ define inverse equivalences of categories

$$\begin{tikzcd}\operatorname{Higgs}_n^d(X) \arrow[r, bend left, "(\pi^*,\pi^\#)"] \arrow[r, phantom, "\cong"] & \operatorname{Higgs}_{n\cdot w(0)}^d(\mathcal{X}), \arrow[l, bend left, "(\pi_*,\pi_\#)"]\end{tikzcd} \tag{2.4}$$

which preserve (semi)stability.

Proof. At the level of vector bundles this is Proposition 1.9. We need to show that the same holds for the Higgs fields. Indeed, computing the two compositions we get

$$\varphi \overset{\pi^\#}{\longmapsto} n\,\pi^*\varphi \otimes s^{n-1} \overset{\pi_\#}{\longmapsto} \frac{1}{n}\,\pi_*(n\,\pi^*\varphi \otimes s^{n-1}) = \varphi$$
$$\eta \overset{\pi_\#}{\longmapsto} \frac{1}{n}\,\pi_*\eta \overset{\pi^\#}{\longmapsto} n\,\pi^*(\frac{1}{n}\,\pi_*\eta) \otimes s^{n-1} = n\,\pi^*\psi \otimes s^{n-1} = \eta,$$

which proves the first statement. As in Proposition 1.9, (semi)stability is preserved trivially. □

The Dolbeault moduli space is defined to be the moduli stack parametrising semistable Higgs bundles, i.e.

$$\mathcal{M}^d_{n,Dol}(X) := \mathrm{Higgs}^{d,ss}_n(X), \quad \mathcal{M}^d_{\underline{m},Dol}(\mathcal{X}) := \mathrm{Higgs}^{d,ss}_{\underline{m}}(\mathcal{X}).$$

After twisting (2.4) with the tautological line bundle on $\mathcal{X}$, we get a version of Theorem 1.10 for the Dolbeault moduli space.

Theorem 2.4. The *twisted* pullback and pushforward functors, defined by

$$\mathrm{Higgs}^d_n(X) \underset{(\pi_*,\pi^\#)(\mathcal{O}(-\frac{d}{n}p)\otimes -)}{\overset{(\pi^*,\pi^\#)(-)\otimes\mathcal{O}(-\frac{d}{n}p)}{\rightleftarrows}} \mathrm{Higgs}^0_{n\cdot w(-d)}(\mathcal{X}).$$

define inverse equivalences of algebraic stacks. These equivalences preserve (semi)stability, hence they restrict to equivalences

$$\mathcal{M}^d_{n,Dol}(X) \underset{(\pi_*,\pi^\#)(\mathcal{O}(-\frac{d}{n}p)\otimes -)}{\overset{(\pi^*,\pi^\#)(-)\otimes\mathcal{O}(-\frac{d}{n}p)}{\rightleftarrows}} \mathcal{M}^0_{n\cdot w(-d),Dol}(\mathcal{X}).$$

Proof. This is a straightforward consequence of Proposition 2.3, obtained with the same argument as discussed in Theorem 1.10. □

3 De Rham and Betti moduli spaces

In the previous section we considered the Dolbeault moduli space of the stacky curve $\mathcal{X}$. Keeping in mind the end goal of studying the non abelian Hodge correspondence on $\mathcal{X}$, we now come to the study of the two other moduli spaces appearing in the correspondence, namely the De Rham and Betti moduli spaces. These are defined as follows

$$\mathcal{M}_{n,DR}(\mathcal{X}) := \langle(\mathcal{E},\nabla) \mid \mathcal{E} \in \mathcal{B}un_n(\mathcal{X}),\ \nabla\colon \mathcal{E} \to \mathcal{E}\otimes\Omega_X \text{ is a flat connection}\rangle$$
$$\mathcal{M}_{n,B}(\mathcal{X}) := [\mathrm{Hom}(\pi_1(\mathcal{X}), GL_n)/GL_n].$$

The Riemann-Hilbert correspondence establishes a natural isomorphism in the analytic topology

$$\mathcal{M}_{n,DR}(\mathcal{X})^{\mathrm{an}} \cong \mathcal{M}_{n,B}(\mathcal{X})^{\mathrm{an}}.$$

As already observed for the moduli space of vector bundles and for the Dolbeault moduli space, also the de Rham moduli space admits a decomposition in connected components

$$\mathcal{M}_{n,DR}(\mathcal{X}) = \coprod_{(\underline{m},d)} \mathcal{M}^d_{\underline{m},DR}(\mathcal{X}),$$

with the latter union ranging over the pairs $(\underline{m} = m_0 w(0) + \dots m_{n-1} w(n-1),\ d)$ such that $\sum_{i=0}^{n-1} m_i = n$. Recall that for a vector bundle to admit a flat connection it is a necessary condition that $\deg \mathcal{E} = 0$. Therefore we get

$$\mathcal{M}_{n,DR}(\mathcal{X}) = \coprod_{\underline{m}} \mathcal{M}^0_{\underline{m},DR}(\mathcal{X}). \tag{3.1}$$

The connected component that we will need to consider is $\mathcal{M}^0_{n\cdot w(-d),\,DR}(\mathcal{X})$. We now want to understand what this corresponds to on the Betti side.

3.1 The fundamental group

In order to decompose the Betti moduli space in connected components, we first need to find an explicit expression for the fundamental group $\pi_1(\mathcal{X})$.

Fix a point $x \in \mathcal{X} \setminus \{p\}$. By $\pi_1(\mathcal{X}, x)$ we mean the fundamental group of the complex analytic orbifold underlying the stacky curve $\mathcal{X}$. This is equivalent to considering $\mathcal{X}$ as a topological stack with respect to the analytic topology and taking Noohi's fundamental group, defined in [Noo05].

Since $\mathcal{X}$ is connected, locally path connected and semilocally 1-connected, following Noohi's approach we get that $\pi_1(\mathcal{X}, x)$ can be computed from the covering spaces of $\mathcal{X}$, as the automorphism group of the fiber functor over x, see [Noo05, pp. 18.13, 18.19, 18.22]. This is analogous to Grothendieck's theory of Galois categories, presented in [Gro71].

The advantage of computing the fundamental group using covering spaces is that we easily get a version of the Seifert-Van Kampen theorem. This is completely analogous to the classical setting of Galois categories, as exposed for example in [DD20, Theorem 4.7.7]. However, since an explicit version of this theorem in our setting seems to be missing, we report the proof for completeness.

Theorem 3.1. Let $\mathcal{X}$ be a topological stack and suppose that it is connected, locally path connected and semilocally 1-connected. Let $\mathcal{U}_1 \overset{i_1}{\hookrightarrow} \mathcal{X}$ and $\mathcal{U}_2 \overset{i_2}{\hookrightarrow} \mathcal{X}$ be open substacks such that $\mathcal{X} = \mathcal{U}_1 \cup \mathcal{U}_2$. Let $\mathcal{V} := \mathcal{U}_1 \times_{\mathcal{X}} \mathcal{U}_2$. Suppose that $\mathcal{U}_1, \mathcal{U}_2$ and $\mathcal{V}$ are connected. Then, for $x \in \mathcal{V}$, there is an isomorphism

$$\pi_1(\mathcal{X}, x) \cong \pi_1(\mathcal{U}_1, x) *_{\pi_1(\mathcal{V},x)} \pi_1(\mathcal{U}_2, x).$$

Equivalently, the following diagram is cocartesian

$$\begin{tikzcd}
\pi_1(\mathcal{V}, x) \arrow[r] \arrow[d] & \pi_1(\mathcal{U}_1, x) \arrow[d] \\
\pi_1(\mathcal{U}_2, x) \arrow[r] \arrow[ur, phantom, "\ulcorner"] & \pi_1(\mathcal{X}, x).
\end{tikzcd}$$

Proof. In order to simplify the notation, let $G := \pi_1(\mathcal{X}, x)$, $H := \pi_1(\mathcal{V}, x)$ and $G_l := \pi_1(\mathcal{U}_l, x)$ for $l = 1, 2,$. The inclusions

$$\begin{tikzcd}
\mathcal{V} \arrow[r, hook, "j_l"] \arrow[rr, hook, bend left, "j"] & \mathcal{U}_l \arrow[r, hook, "i_l"] & \mathcal{X}
\end{tikzcd}$$

give rise to a commutative diagram

$$\begin{tikzcd}
H \arrow[r, "j_{1*}"] \arrow[d, "j_{2*}"'] & G_1 \arrow[d, "\iota_1"] \arrow[ddr, bend left, "i_{1*}"] & \\
G_2 \arrow[r, "\iota_2"] \arrow[drr, bend right, "i_{2*}"'] & G_1 *_H G_2 \arrow[dr, dashed, "v"] & \\
& & G,
\end{tikzcd}$$

where the dashed arrow can be filled out by the universal property of $G_1 *_H G_2$. We want to prove that v is an isomorphism.

Consider the following 2-commutative diagram

$$\begin{tikzcd}
\mathbf{Cov}_{\mathcal{X}} \arrow[rr, "\theta = F_x^{\mathcal{X}}"] \arrow[d, "\alpha"] & & G\text{-}\mathbf{Set} \arrow[d, "\beta"] \\
\mathbf{Cov}_{\mathcal{U}_1} \times_{\mathbf{Cov}_{\mathcal{V}}} \mathbf{Cov}_{\mathcal{U}_2} \arrow[rr, "\varphi = F_x^{\mathcal{U}_1} \times_{F_x^{\mathcal{V}}} F_x^{\mathcal{U}_2}"] & & G_1\text{-}\mathbf{Set} \times_{H\text{-}\mathbf{Set}} G_2\text{-}\mathbf{Set},
\end{tikzcd}$$

where α is induced by the restriction of covering spaces and β is induced by i_{1*} and i_{2*}. Notice that α is an equivalence of categories, obtained by the glueing of covering spaces, φ and θ are equivalences, as a consequence of [Noo05, Theorem 18.19]. Therefore, β is an equivalence as well.

Notice moreover that the homomorphisms (ι_1, ι_2) induce an equivalence of categories

$$\gamma \colon (G_1 *_H G_2)\text{-}\mathbf{Set} \to G_1\text{-}\mathbf{Set} \times_{H\text{-}\mathbf{Set}} G_2\text{-}\mathbf{Set},$$

making the following diagram 2-commute

$$\begin{tikzcd}
G\text{-}\mathbf{Set} \arrow[dr, "v^*"] \arrow[dd, "\beta"'] & \\
& (G_1 *_H G_2)\text{-}\mathbf{Set} \arrow[dl, "\gamma"] \\
G_1\text{-}\mathbf{Set} \times_{H-set} G_2\text{-}\mathbf{Set}. &
\end{tikzcd}$$

Since both β and γ are equivalences, this shows that v^* is also an equivalence and therefore, see [DD20, Proposition 4.7.1], that v is an isomorphism of groups. □

Using this theorem we can compute the fundamental group of the stacky curve $\mathcal{X}$.

Proposition 3.2. The fundamental group of the stacky curve $\mathcal{X}$ is

$$\pi_1(\mathcal{X}) = \left\langle a_i,\ b_i,\ c \;\middle|\; \prod_{i=1}^{g} [a_i,\ b_i] = c,\ c^n = 1 \right\rangle.$$

Proof. As mentioned in Section 1, if $U \cong \mathbb{A}^1$ is an affine neighborhood of p in X, then

$$\mathcal{X}|_U \cong [\mathbb{A}^1/\mu_n].$$

Let then $\mathcal{U}_1 := \mathcal{X}|_U = \pi^{-1}(U), \mathcal{U}_2 := \mathcal{X} \setminus \{p\}$ and take a point $x \in \mathcal{U}_1 \cap \mathcal{U}_2 =: \mathcal{V}$. Then by (1.3) the substack $\mathcal{U}_2$ is a punctured curve and therefore

$$\pi_1(\mathcal{U}_2, x) = \langle a_i, b_i \mid i = 1, \dots g \rangle.$$

In order to compute $\pi_1(\mathcal{U}_1, x)$, notice that $[\mathbb{A}^1/\mu_n]$ is a Deligne-Mumford stack with atlas $q\colon \mathbb{A}^1 \to [\mathbb{A}^1/\mu_n]$. In particular, the morphism q is étale, i.e. a covering map. Since $\mathbb{A}^1$ is simply connected, this is the universal cover of $\mathcal{U}_1$. By [Noo05, Corollary 18.20] the fundamental group can be identified with the group of deck transformations of the universal cover. Since q is a μ_n-bundle, this shows that

$$\pi_1(\mathcal{U}_1, x) = \mu_n = \langle c \mid c^n = 1 \rangle.$$

Finally, the intersection $\mathcal{V}$ is isomorphic to $\mathbb{A}^1 \setminus \{0\}$, therefore

$$\pi_1(\mathcal{V}, x) = \langle d \rangle.$$

Now we can use Theorem 3.1 to compute the fundamental group of $\mathcal{X}$. Indeed, the inclusions $j_1\colon \mathcal{V} \to \mathcal{U}_1$ and $j_2\colon \mathcal{V} \to \mathcal{U}_2$ induce

$$\begin{aligned} j_{1*}\colon \pi_1(\mathcal{V}, x) &\to \pi_1(\mathcal{U}_1, x) \\ d &\mapsto c, \end{aligned}$$

$$\begin{aligned} j_{2*}\colon \pi_1(\mathcal{V}, x) &\to \pi_1(\mathcal{U}_2, x) \\ d &\mapsto \prod_{i=1}^{g} [a_i, b_i]. \end{aligned}$$

Taking the pushout we get the above presentation of $\pi_1(\mathcal{X})$. □

Corollary 3.3. As a consequence, we get an explicit description of the Betti moduli space

$$\mathcal{M}_{n,B}(\mathcal{X}) \cong \left[\left\{ ((A_i, B_i)_{i=1}^{g}, C) \in GL_n^{2g+1} \;\middle|\; \prod_{i=1}^{g} [A_i, B_i] = C,\, C^n = Id \right\} / GL_n \right]. \tag{3.2}$$

3.2 The monodromy representation

We now want to understand what the component $\mathcal{M}^0_{n \cdot w(-d),DR}(\mathcal{X})$ corresponds to in $\mathcal{M}_{n,B}(\mathcal{X})$ under the isomorphism

$$\mathcal{M}_{n,DR}(\mathcal{X})^{\mathrm{an}} \cong \mathcal{M}_{n,B}(\mathcal{X})^{\mathrm{an}}, \tag{3.3}$$

according to the description of $\mathcal{M}_{n,B}(\mathcal{X})$ given in Corollary 3.3.

The isomorphism in Equation (3.3) is obtained by mapping a pair $(\mathcal{E}, \nabla)$ to the associated monodromy representation at x. In particular, the matrices A_i, B_i, C in (3.2) are obtained by the action of the generators a_i, b_i, c of the fundamental group of $\mathcal{X}$, as expressed in Proposition 3.2, and are well defined up to conjugation.

The multiplicity vector $\underline{m}(\mathcal{E})$ depends only on the behavior of $\mathcal{E}$ at the stacky point p, and therefore can be computed in a neighborhood of p isomorphic to $[\mathbb{A}^1/\mu_n]$. In particular, looking back at the Seifert-Van Kampen computation of $\pi_1(\mathcal{X}, x)$ in Proposition 3.2, we see that the decomposition (3.1) will translate on the Betti moduli space only in terms of conditions on the matrix C.

In particular, after restricting to $[\mathbb{A}^1/\mu_n]$, we can find the matrix C as obtained by the monodromy action of the generator of $\pi_1([\mathbb{A}^1/\mu_n], x) \cong \mu_n$ on $(\mathcal{E}|_{[\mathbb{A}^1/\mu_n]}, \nabla|_{[\mathbb{A}^1/\mu_n]})$.

Proposition 3.4. Let $(\mathcal{E}, \nabla)$ be a vector bundle with a flat connection on $[\mathbb{A}^1/\mu_n]$. Then the representation of $\pi_1([\mathbb{A}^1/\mu_n], x) \cong \mu_n$ obtained by the monodromy action on $(\mathcal{E}|_{[\mathbb{A}^1/\mu_n]}, \nabla|_{[\mathbb{A}^1/\mu_n]})$ coincides with the representation obtained as $\mathcal{E}|_{B\mu_n}$.

Proof. As mentioned in the proof of Proposition 3.2, the projection map

$$p\colon \mathbb{A}^1 \to [\mathbb{A}^1/\mu_n]$$

is a principal μ_n-bundle and it is the universal cover of $[\mathbb{A}^1/\mu_n]$. The datum of the pair $(\mathcal{E}, \nabla)$ is equivalent to the μ_n-equivariant pair $(p^*\mathcal{E}, p^*\nabla)$ on $\mathbb{A}^1$. The fundamental group of $[\mathbb{A}^1/\mu_n]$ is identified with μ_n and under this identification the monodromy action on $(\mathcal{E}, \nabla)$ is the pushforward of the μ_n-action on $(p^*\mathcal{E}, p^*\nabla)$.

Since $\mathbb{A}^1$ is simply connected and $p^*\nabla$ is a flat connection on $p^*\mathcal{E}$, the pair $(p^*\mathcal{E}, p^*\nabla)$ is actually equivalent to the trivial bundle, with the standard derivation. Moreover, since $\mathbb{A}^1$ is affine, this is just the sheafification of the pair $(\mathbb{C}[x]^n, \mathrm{d})$. The μ_n-equivariant structure corresponds to an action

$$\mu_n \to \mathrm{Aut}_{\mathbb{C}[x]}((\mathbb{C}[x]^n, \mathrm{d})).$$

Notice that an automorphism $\varphi\colon \mathbb{C}[x]^n \to \mathbb{C}[x]^n$ fixes the differential d if and only if $\mathrm{d}(\varphi) = 0$. Therefore we have the following isomorphisms

$$\begin{array}{ccc} \mathrm{Aut}_{\mathbb{C}[x]}(\mathbb{C}[x]^n) & \cong & GL_n(\mathbb{C}[x]) \\ \cup\!\mid & & \cup\!\mid \\ \mathrm{Aut}_{\mathbb{C}[x]}((\mathbb{C}[x]^n, \mathrm{d})) & \cong & GL_n(\mathbb{C}). \end{array}$$

In particular, the μ_n-equivariant structure on $(p^*\mathcal{E}, p^*\nabla)$ gives an action

$$\rho\colon \mu_n \to GL_n(\mathbb{C}),$$

i.e. an n-dimensional representation of μ_n, which is exactly the monodromy representation of the pair $(\mathcal{E}, \nabla)$.

We now want to compare this with the representation obtained as $\mathcal{E}|_{B\mu_n}$. Consider

$$\begin{tikzcd} * = \{0\} \arrow[r, hook, "i"] \arrow[d] & \mathbb{A}^1 \arrow[d, "p"] \\ B\mu_n \arrow[r, hook, "j"'] & {[\mathbb{A}^1/\mu_n]}. \end{tikzcd}$$

The representation associated to $\mathcal{E}|_{B\mu_n}$ is obtained by the μ_n-equivariant structure of $(p|_{\{0\}})^*\mathcal{E}|_{B\mu_n}$, corresponding to a morphism

$$\mu_n \to \mathrm{Aut}_{\{0\}}((p|_{\{0\}})^*\mathcal{E}|_{B\mu_n}) = \mathrm{Aut}_{\{0\}}(i^*p^*\mathcal{E}).$$

Under the identifications $\mathrm{QCoh}(\mathbb{A}^1) \cong \mathbb{C}[x]\text{-Mod}$ and $\mathrm{QCoh}(*) \cong \mathbb{C}\text{-Mod}$, the restriction i^* is just the morphism of evaluation at 0. Since $p^*\mathcal{E}$ is the trivial rank n vector bundle, we get

$$\begin{tikzcd}
\mu_n \arrow[r, "\rho'"] \arrow[dr, "\sigma"'] & GL_n(\mathbb{C}[x]) \arrow[d, "\mathrm{ev}_0"] & \cong & \mathrm{Aut}_{\mathbb{A}^1}(p^*\mathcal{E}) \arrow[d, "i^*"] \\
& GL_n(\mathbb{C}) & \cong & \mathrm{Aut}_{\{0\}}(i^*p^*\mathcal{E}).
\end{tikzcd}$$

But ρ' must factor through the inclusion

$$\mathrm{Aut}_{\mathbb{A}^1}((p^*\mathcal{E}, \nabla)) \subseteq \mathrm{Aut}_{\mathbb{A}^1}(p^*\mathcal{E}),$$

so we get

$$\begin{tikzcd}
\mu_n \arrow[r, "\rho"] \arrow[rr, bend left, "\rho'"] \arrow[drr, "\sigma"'] & GL_n(\mathbb{C}) & \subseteq & GL_n(\mathbb{C}[x]) \arrow[d, "\mathrm{ev}_0"] \\
& & & GL_n(\mathbb{C}).
\end{tikzcd}$$

Clearly the composition $GL_n(\mathbb{C}) \subseteq GL_n(\mathbb{C}[x]) \xrightarrow{\mathrm{ev}_0} GL_n(\mathbb{C})$ is the identity, so the representations ρ and σ coincide. $\square$

Corollary 3.5. Under the identification (3.2), the connected component of the Betti moduli space corresponding to the multiplicity vector $\underline{m} = m_0\, w(0) + \cdots + m_{n-1}\, w(n-1)$ is

$$\mathcal{M}_{\underline{m},B}(\mathcal{X}) \cong \left[\left\{ (A_i, B_i)_{i=1}^g \in GL_n^{2g} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = C_{\underline{m}} \right\} / GL_n \right],$$

where $C_{\underline{m}}$ is a matrix representing the action of the generator of μ_n in the representation with multiplicity vector $\underline{m}$. In particular, $C_{\underline{m}}$ can be taken as a diagonal matrix with exactly m_l entries equal to $\mathrm{e}^{\frac{2\pi \mathrm{i}}{n} l}$.

In particular, the case we are interested in is $\underline{m} = n \cdot w(-d)$.

Corollary 3.6. The Riemann-Hilbert correspondence (3.3) maps the weight $-d$ component of the De Rham moduli space $\mathcal{M}_{n\cdot w(-d),DR}(\mathcal{X})$ to

$$\mathcal{M}_{n\cdot w(-d),B}(\mathcal{X}) \cong \left[\left\{ (A_i, B_i)_{i=1}^g \in GL_n^{2g} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = \mathrm{e}^{-\frac{2\pi \mathrm{i}}{n} d} Id \right\} / GL_n \right].$$

4 The non abelian Hodge correspondence

In [Sim11], Simpson showed that the non abelian Hodge correspondence can be extended from smooth projective varieties to smooth and proper Deligne-Mumford stacks. Denote by M_{Dol}, M_{DR} and M_B the coarse moduli spaces of the Dolbeult, De Rham and Betti moduli stacks.

Let $\mathcal{Y}$ be a smooth and proper DM stack. In [Sim11], Simpson proved the Riemann-Hilbert isomorphism

$$M^0_{n,DR}(\mathcal{Y})^{\mathrm{an}} \cong M_{n,B}(\mathcal{Y})^{\mathrm{an}}, \tag{4.1}$$

and the non-abelian Hodge homeomorphism

$$M^0_{n,Dol}(\mathcal{Y})^{\mathrm{top}} \cong M^0_{n,DR}(\mathcal{Y})^{\mathrm{top}}. \tag{4.2}$$

Let $\mathcal{Y} = \mathcal{X}$ be the stacky curve that we have studied so far. Of course, (4.1) and (4.2) respect the decomposition in connected components, so they restrict to

$$M^0_{\underline{m},DR}(\mathcal{X})^{\mathrm{an}} \cong M_{\underline{m},B}(\mathcal{Y})^{\mathrm{an}},$$
$$M^0_{\underline{m},Dol}(\mathcal{X})^{\mathrm{top}} \cong M^0_{\underline{m},DR}(\mathcal{X})^{\mathrm{top}}.$$

By Corollary 3.3, the coarse moduli space $M_{\underline{m},B}(\mathcal{X})$ can be described as a GIT quotient

$$M_{\underline{m},B}(\mathcal{X}) = \left\{ (A_i, B_i)_{i=1}^g \in GL_n^{2g} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = C_{\underline{m}} \right\} //GL_n$$

and in particular

$$M_{n\cdot w(-d),B}(\mathcal{X}) = \left\{ (A_i, B_i)_{i=1}^g \in GL_n^{2g} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = \mathrm{e}^{-\frac{2\pi \mathrm{i}}{n} d} Id \right\} //GL_n.$$

On the other hand, the isomorphism proven in Theorem 2.4 induces an isomorphism of the coarse moduli spaces

$$M^d_{n,Dol}(X) \cong M^0_{n\cdot w(-d),Dol}(\mathcal{X}).$$

Proposition 4.1. Putting the above homeomorphisms together, we get

$$M^d_{n,Dol}(X)^{\mathrm{top}} \cong M^0_{n\cdot w(-d),DR}(\mathcal{X})^{\mathrm{top}} \cong \left(\left\{ (A_i, B_i)_{i=1}^g \in GL_n^{2g+1} \;\middle|\; \prod_{i=1}^g [A_i, B_i] = \mathrm{e}^{-\frac{2\pi \mathrm{i}}{n} d} Id \right\} //GL_n \right)^{\mathrm{top}}.$$

This is exactly the non abelian Hodge correspondence for vector bundles of degree d on X, where we gave a description of the twisted character variety of X as a component of the character variety of $\mathcal{X}$.

Of course these homeomorphisms induce isomorphisms in singular cohomology

$$H^*(M^d_{n,Dol}(X)) = H^*(M^d_{n\cdot w(-d),Dol}(\mathcal{X})) \cong H^*(M^0_{n\cdot w(-d),DR}(\mathcal{X})) \cong H^*(M_{n\cdot w(-d),B}(\mathcal{X})).$$

We now give a more explicit description of these isomorphisms, with the goal of understanding how they relate the Hodge structures. The key idea of this approach, due to Simpson, is that the non abelian Hodge correspondence can be understanding as relating the geometry of the generic and the special fiber inside the Hodge moduli space, i.e. the moduli space of vector bundles with λ-connections.

Definition 4.2. Let $\mathcal{E}$ be a vector bundle on $\mathcal{X}$. For a fixed complex number λ, define a *λ-connection* $\mathcal{E}$ to be a $\mathbb{C}$-linear morphism $\nabla\colon \mathcal{E} \to \mathcal{E} \otimes \Omega_{\mathcal{X}}$, respecting the Leibniz rule up to the scalar λ, namely

$$\nabla(f \cdot s) = s \otimes \mathrm{d}f + \lambda f \cdot \nabla(s)$$

for all sections $f \in \mathcal{O}_{\mathcal{X}}(U)$ and $s \in \mathcal{E}(U)$. Define the *Hodge moduli space* of $\mathcal{X}$ to be the moduli stack parametrizing semistable triples $(\mathcal{E}, \lambda, \nabla)$, where $\mathcal{E}$ is a vector bundle on $\mathcal{X}$, ∇ is a λ-connection on $\mathcal{E}$ and $\mathcal{E}$ is slope-semistable with respect to ∇. Explicitly,

$$\mathcal{M}^0_{\underline{m},Hdg}(\mathcal{X}) := \left\langle (\mathcal{E}, \lambda, \nabla) \;\middle|\; \begin{array}{c} \mathcal{E} \in \mathcal{B}un^0_{\underline{m}}(\mathcal{X}),\ \lambda \in \mathbb{A}^1,\ \nabla\colon \mathcal{E} \to \mathcal{E} \otimes \Omega_X \text{ is a } \lambda - \text{connection}, \\ (\mathcal{E}, \nabla) \text{ is semistable} \end{array} \right\rangle.$$

Remark 4.3. We have the following properties.

- The multiplicative group $\mathbb{G}_m$ acts on $\mathcal{M}^0_{\underline{m},Hdg}(\mathcal{X})$ by

$$t \cdot (\mathcal{E}, \nabla, \lambda) = (\mathcal{E}, t\nabla, t\lambda).$$

- The projection

$$\begin{aligned} \pi\colon \mathcal{M}^0_{\underline{m},Hdg}(\mathcal{X}) &\to \mathbb{A}^1 \\ (\mathcal{E}, \nabla, \lambda) &\mapsto \lambda, \end{aligned}$$

  is $\mathbb{G}_m$-equivariant, where $\mathbb{G}_m$ acts on $\mathbb{A}^1$ with weight one.

- For $\lambda \neq 0$ all of the fibers $\pi^{-1}(\lambda)$ are isomorphic to $\pi^{-1}(1) = \mathcal{M}^0_{\underline{m},DR}(\mathcal{X})$, while $\pi^{-1}(0) = \mathcal{M}^0_{\underline{m},Dol}(\mathcal{X})$.

Summing up, we have the following diagram

$$\begin{tikzcd} \mathcal{M}^0_{\underline{m},Dol}(\mathcal{X}) \arrow[r, hook, "i"] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & \mathcal{M}^0_{\underline{m},Hdg}(\mathcal{X}) \arrow[d, "\pi"] & \mathcal{M}^0_{\underline{m},DR}(\mathcal{X}) \arrow[l, hook', "j"'] \arrow[r, "\psi", "\cong"'] \arrow[d] \arrow[dl, phantom, "\llcorner"] & \mathcal{M}_{\underline{m},B}(\mathcal{X}) \\ \{0\} \arrow[r] & \mathbb{A}^1 & \{1\}, \arrow[l] & \end{tikzcd} \tag{4.3}$$

where ψ is defined analytically via the Riemann-Hilbert correspondence and π is a trivial fibration over $\mathbb{A}^1 \setminus \{0\}$.

Restrict now to the case $\underline{m} = n \cdot w(-d)$ and suppose that n and d are coprime. This hypothesis ensures that in a point $(\mathcal{E}, \lambda, \nabla)$ is semistable if and and only if it is stable. Consequently, the coarse moduli space $M^0_{n \cdot w(-d),Hdg}(\mathcal{X})$ is smooth, the morphism π and the moduli map

$$\mathcal{M}^0_{n \cdot w(-d),Hdg}(\mathcal{X}) \to M^0_{n \cdot w(-d),Hdg}(\mathcal{X})$$

is a $\mathbb{G}_m$-gerbe.

Descending the diagram (4.3) to the coarse moduli spaces we get a diagram of smooth varieties

$$\begin{tikzcd}
M^0_{n\cdot w(-d),Dol}(\mathcal{X}) \arrow[r, hook, "i"] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & M^0_{n\cdot w(-d),Hdg}(\mathcal{X}) \arrow[d, "\pi"] & M^0_{n\cdot w(-d),DR}(\mathcal{X}) \arrow[l, hook', "j"'] \arrow[r, "\psi", "\cong"'] \arrow[d] \arrow[dl, phantom, "\llcorner"] & M_{n\cdot w(-d),B}(\mathcal{X}) \\
\{0\} \arrow[r] & \mathbb{A}^1 & \{1\}, \arrow[l] &
\end{tikzcd}$$

Using the $\mathbb{G}_m$-action, we can use a Białynicki-Birula decomposition of $M^0_{n\cdot w(-d),Hdg}(\mathcal{X})$ to show that i and j induce isomorphisms in cohomology. The idea to use the Białynicki-Birula decomposition to make this cohomological computation was inspired by by Nakajima's appendix to [CV04].

Theorem 4.4. The inclusions i and j induce isomorphisms of pure mixed Hodge structures in cohomology.

Proof. The Hodge moduli space $M^0_{n\cdot w(-d),Hdg}(\mathcal{X})$ is a *semiprojective variety*, according to the terminology used in [HR15]. This means that the fixed-point locus

$$M^0_{n\cdot w(-d),Hdg}(\mathcal{X})^{\mathbb{G}_m}$$

is proper and that the limit point $\lim_{t\to 0} t\cdot x$ exists for all $x \in M^0_{n\cdot w(-d),Hdg}(\mathcal{X})$. Indeed, these properties are [Sim97, Lemma 10.4, Corollary 10.3]. Therefore the theorem is just [HR15, Corollary 1.3.3]. □

Remark 4.5. The proof of Theorem 4.4 fails to be algebraic in two different ways. On one side, to show that $M^0_{n\cdot w(-d),Hdg}(\mathcal{X})$ is a semiprojective variety we employed Simpsons' results [Sim97, Lemma 10.4, Corollary 10.3], which are proved analytically. On the other side, the proof of [HR15, Corollary 1.3.3] is itself analytic. In the appendix we solve the second issue by giving an algebraic proof of [HR15, Corollary 1.3.3].

Appendix

We give an algebraic proof of Hausel and Rodriguez-Villega's result [HR15, Corollary 1.3.3].

Theorem 4.6. Let M be a smooth semiprojective variety and let $\pi\colon M \to \mathbb{A}^1$ be a $\mathbb{G}_m$-equivariant morphism, where $\mathbb{G}_m$ acts on $\mathbb{A}^1$ with weight 1. Denote by $M_t \subseteq M$ the fiber over $t \in \mathbb{A}^1$. Consider the inclusions

$$\begin{tikzcd} M_0 \arrow[r, hook, "i_0"] \arrow[d] \arrow[dr, phantom, "\lrcorner"] & M \arrow[d, "\pi"] & M_1 \arrow[l, hook', "i_1"'] \arrow[d] \arrow[dl, phantom, "\llcorner"] \\ \{0\} \arrow[r] & \mathbb{A}^1 & \{1\}. \arrow[l] \end{tikzcd}$$

Then i_0 and i_1 induce isomorphisms in cohomology.

Proof. The fact that i_0 induces an isomorphism in cohomology is standard. Indeed, thanks to [HR15, Eq. 1.2.2], the Białynicki-Birula decomposition of M gives a stratification which can be used to compute the cohomology. Since the $\mathbb{G}_m$-action restricts to M_0, the stratification restricts to the Białynicki-Birula stratification of M_0. Using these stratifications to compute the singular cohomology, one can easily show that i_0 induces an isomorphism in cohomology.

In order to show that also i_1 induces an isomorphism in cohomology, it is convenient to factor it as

$$M_1 \hookrightarrow M_{\neq 0} \overset{j}{\hookrightarrow} M,$$

where $M_{\neq 0}$ is the fiber over $\mathbb{G}_m = \mathbb{A}^1 \setminus \{0\}$. Since $M_{\neq 0}$ is open in M with complement M_0, the diagram

$$M_0 \overset{i_0}{\hookrightarrow} M \overset{j}{\hookleftarrow} M_{\neq 0}$$

yields a short exact sequence of (constructible) sheaves of $\mathbb{Q}$-vector spaces on M

$$0 \to j_!\underline{\mathbb{Q}}_{M\neq 0} \to \underline{\mathbb{Q}}_M \to i_{0,*}\underline{\mathbb{Q}}_{M_0} \to 0. \tag{4.4}$$

[add ref, i.e. Vakil, this is often referred to as the gysin sequence, we are taking the etale topology]

Let n be the topological dimension of M. The dualizing sheaf for Verdier duality is $\underline{\mathbb{Q}}_M[n]$. Namely, if $\mathcal{F}$ is a constructible sheaf of $\mathbb{Q}$-vector spaces on M, then its Verider dual in the derived category is $\mathbb{D}_M\mathcal{F} = \mathbf{R}\mathcal{H}om(\mathcal{F}, \underline{\mathbb{Q}}_M[n])$. In the case of the sheaves appearing in the sequence (4.4) we get the following identifications.

$$\begin{aligned} \mathbb{D}_M j_!\underline{\mathbb{Q}}_{M_{\neq 0}} &= \mathbf{R}\mathcal{H}om(j_!\underline{\mathbb{Q}}_{M_{\neq 0}}, \underline{\mathbb{Q}}_M[n]) \\ &= \mathbf{R}\mathcal{H}om(\mathbf{R}j_!\underline{\mathbb{Q}}_{M_{\neq 0}}, \underline{\mathbb{Q}}_M[n]) && (4.5) \\ &= \mathbf{R}j_*\mathbf{R}\mathcal{H}om(\underline{\mathbb{Q}}_{M_{\neq 0}}, j^!\underline{\mathbb{Q}}_M[n]) && (4.6) \\ &= \mathbf{R}j_*\mathbf{R}\mathcal{H}om(\underline{\mathbb{Q}}_{M_{\neq 0}}, \underline{\mathbb{Q}}_{M_{\neq 0}}[n]) && (4.7) \\ &= \mathbf{R}j_*\mathbf{R}\mathcal{H}om(\underline{\mathbb{Q}}_{M_{\neq 0}}, \underline{\mathbb{Q}}_{M_{\neq 0}})[n] \\ &= \mathbf{R}j_*\underline{\mathbb{Q}}_{M_{\neq 0}}[n], \end{aligned}$$

where (4.5) is due to $j_!$ being exact since j is an open immersion; (4.6) is Verdier duality; and (4.7) is due to the fact that $M_{\neq 0}$ has the same dimension as M and $j^!$ takes the dualizing sheaf on M to the dualizing sheaf on $M_{\neq 0}$.
By definition $\mathbb{D}_M \underline{\mathbb{Q}}_M = \underline{\mathbb{Q}}_M[n]$ and finally

$$\begin{aligned} \mathbb{D}_M i_{0,*}\underline{\mathbb{Q}}_{M_0} &= i_{0,!}\mathbb{D}_{M_0}\underline{\mathbb{Q}}_{M_0} && (4.8)\\ &= i_{0,!}\underline{\mathbb{Q}}_{M_0}[n-2] && (4.9)\\ &= i_{0,*}\underline{\mathbb{Q}}_{M_0}[n-2], && (4.10) \end{aligned}$$

where (4.8) is a property of Verdier duality, (4.9) is because the topological dimension of M_0 is $n-2$ and finally (4.10) is due to i being a closed immersion.

Therefore, applying the Verdier duality functor to the Gysin sequence (4.4), we get the exact triangle

$$i_*\underline{\mathbb{Q}}_{M_0}[-2] \to \underline{\mathbb{Q}}_M \to \mathbf{R}j_*\underline{\mathbb{Q}}_{M_{\neq 0}}.$$

Summing up, we get a commutative diagram

$$\begin{tikzcd} i_*\underline{\mathbb{Q}}_{M_0}[-2] \arrow[r] \arrow[dr] & \underline{\mathbb{Q}}_M \arrow[r] \arrow[d] & \mathbf{R}j_*\underline{\mathbb{Q}}_{M_{\neq 0}} \\ & i_*\underline{\mathbb{Q}}_{M_0}, & \end{tikzcd}$$

where the horizontal line is an exact triangle and the vertical morphism induces isomorphisms in cohomology.
Applying the functor $\mathbf{R}\Gamma(M,-)$, we get

$$\begin{tikzcd} H^{*-2}(M_0,\mathbb{Q}) \arrow[r, "(\star\star)"] \arrow[dr, "(\star)"'] & H^*(M,\mathbb{Q}) \arrow[r] \arrow[d, "\cong"] & H^*(M_{\neq 0},\mathbb{Q}) \\ & H^*(M_0,\mathbb{Q}) & \end{tikzcd} \tag{4.11}$$

The morphism $(\star)$ is obtained by cup product with $c_1(\mathcal{N})$, i.e. the first Chern class of the normal bundle to M_0 in M. [**TO DO**: add ref, use gysin sequence]
Since π is smooth, $\mathcal{N}$ is the pullback of the normal bundle to to $\{0\}$ in $\mathbb{A}^1$, which is trivial. Therefore $\mathcal{N} \cong \mathcal{O}_{M_0}$ and $c_1(\mathcal{N}) = 0$. This shows that $(\star)$ is the zero map. Substituting in (4.11) we get that $(\star\star)$ is also the zero map. Therefore, we get short exact sequences

$$0 \to H^*(M,\mathbb{Q}) \to H^*(M_{\neq 0},\mathbb{Q}) \to H^{*-1}(M_0,\mathbb{Q}) \to 0. \tag{4.12}$$

Finally, we can use these short exact sequences to show $H^*(M,\mathbb{Q}) \cong H^*(M_1,\mathbb{Q})$. The morphism $\pi\colon M_{\neq 0} \to \mathbb{G}_m$ is a trivial fibration, therefore $M_{\neq 0} \cong M_1 \times \mathbb{G}_m$. By Künneth's formula

$$H^*(M_{\neq 0}) = H^*(M_1) \otimes H^*(\mathbb{G}_m). \tag{4.13}$$

The multiplicative group $\mathbb{G}_m$ has the homotopy type of a circle, so Equation (4.13) becomes

$$H^*(M_{\neq 0}) = H^*(M_1) \oplus H^{*-1}(M_1).$$

Substituting in (4.12) we get short exact sequences

$$0 \to H^*(M) \to H^*(M_1) \oplus H^{*-1}(M_1) \to H^{*-1}(M) \to 0.$$

Then the thesis is proven by induction on the cohomological degree. □

Remark 4.7. Since the inclusions i_0 and i_1 are algebraic, the pullbacks

$$i_0^* \colon H^*(M, \mathbb{Q}) \to H^*(M_0, \mathbb{Q}), \quad i_1^* \colon H^*(M, \mathbb{Q}) \to H^*(M_1, \mathbb{Q})$$

are isomorphisms of mixed Hodge structures. Moreover, M has pure Hodge structure, as shown in [HR15, Corollary 1.3.2]. Therefore i_0^* and i_1^* are isomorphisms of pure mixed Hodge structures.

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